Local Noetherianity of the actual weighted-Proj curve

Formal proof packet

28 October 2026

This packet reports checked statements and open proof obligations. It makes no DLP trial, cost, or attack claim. The curve is $X = \mathrm{Proj}(\mathrm{degreeSubmodule}(a_3, a_2, a_1, a_0))$ over a field K.

Checked declarations

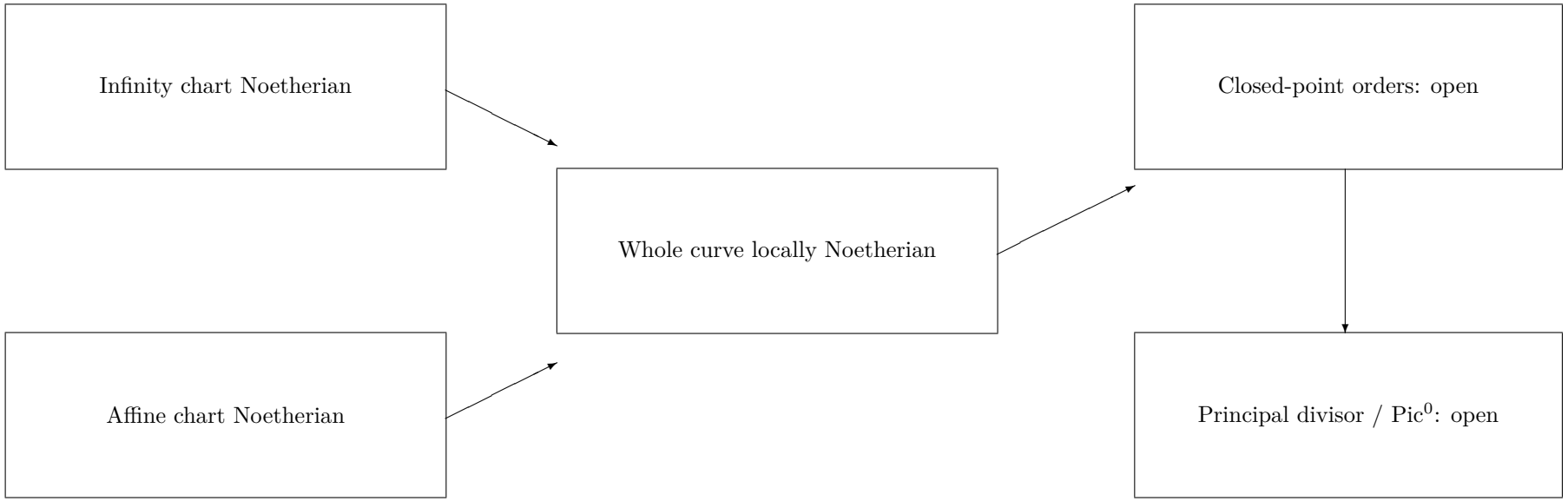
- The actual infinity-chart ring and affine-chart ring are Noetherian. Each is a quotient of a two-variable polynomial ring over K.
- Their scheme isomorphisms transport local Noetherianity to the two actual basic opens. The existing two-open cover then proves $\mathrm{IsLocallyNoetherian}(X)$ for every field and coefficient tuple, without a separability hypothesis.
- Pinned Lean 4.34.1 printed only `propext`, `Classical.choice`, and `Quot.sound` for the three new declarations. The source and direct compiler receipt are retained beside this report.

Scope boundary

The earlier whole-curve integrality theorem separately assumes separability of the quintic. Together these two results supply Mathlib's typeclass prerequisites for order-of-vanishing machinery, but this packet does not identify every closed support point as coheight one, allocate orders of the source function, prove finite support or an infinity pole, construct a principal divisor, prove a Picard-class equality, or link the proof to compiled Rust. The accompanying CSV marks each step individually.

Dependency diagram

Blue in the accompanying SVG denotes checked declarations; gray denotes open proof gates. The diagram below records dependencies, not measured performance.



Existing chart isomorphisms and two-open cover are used; the global divisor and runtime bridge remain open.